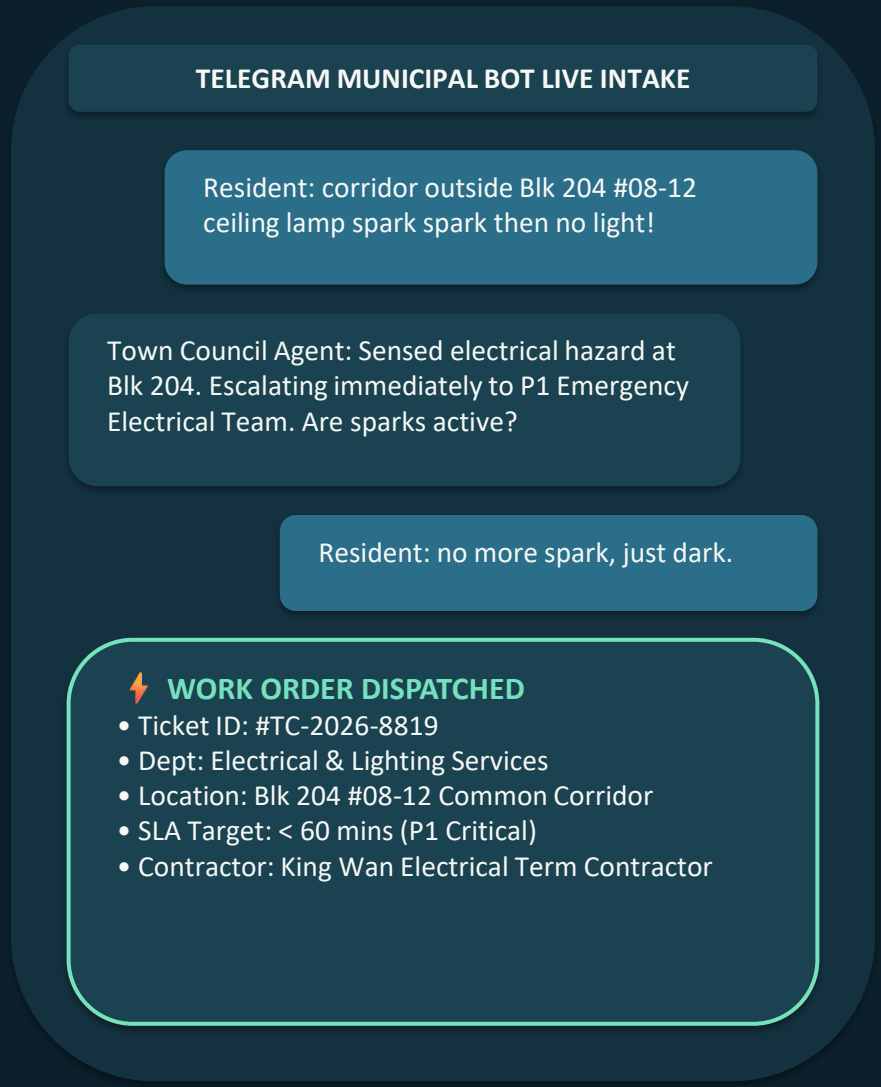
SINGAPORE MUNICIPAL SERVICES

Town Council AI Agent
& Resident Services Chatbot

An autonomous, production-grade multi-agent operations platform for municipal resident inquiries, deterministic emergency triage, by-law enforcement, and automated contractor dispatch across town council precincts.

🏛️ **Core Platform Capabilities:**
- 24/7 Multimodal Resident Intake (Telegram, Audio Voice Memos, Damage Photos)
- Deterministic Emergency Triage (Zero-Latency Safety Bypass for Gas, Fire, Lift Hazards)
- Grounded Semantic RAG for Town Council By-Laws & Contractor Auto-Dispatch



The Municipal Estate Management Challenge

SCALE OF TOWN COUNCIL PRECINCTS

- **High Volume Influx:** Town Councils manage 50,000 to 150,000 residents across hundreds of blocks, receiving thousands of feedback items weekly.
- **Multi-Channel Chaos:** Reports flood in via OneService, phone hotlines, counter walk-ins, and Telegram, creating manual clerk transcription backlogs.
- **Singlish & Dialect Dialects:** Residents submit colloquial, unstructured reports ('chute chok liao', 'void deck ceiling drip water') that standard chatbots fail to parse.
- **Complex Municipal By-Laws:** Dozens of statutory rules governing common corridors, bulky item removal, void deck events, and renovation permits are hard for junior staff to recall quickly.

CRITICAL OPERATIONAL PITFALLS

- **The 'Emergency in Queue' Trap:** Standard LLM chatbots and static ticketing queues treat a gas leak or electrical sparking the same as a paint chip—causing dangerous dispatch delays.
- **LLM Hallucinations on Tariffs:** Generic chatbots invent fake fees or wrong hotline numbers, creating severe public relations and legal liability for the Town Council.
- **Endless Clarification Loops:** Poorly designed agents interrogate residents with 5-10 back-and-forth questions, leading to high resident abandonment.
- **Zero Downstream Integration:** Most chatbots stop at answering questions; they do not extract structured location data, calculate SLAs, or dispatch contractors automatically.

🎯 **System Goal: A deterministic-first, multimodal agentic platform that guarantees zero emergency slip-through, accurately grounds municipal by-laws via Semantic RAG, and automates contractor work orders.**

Multi-Agent System Architecture

1. INTAKE & SENSING

Telegram Bot
Residents message text, photos, or voice notes 24/7.

Voice Memo AI
Whisper / Gemini Voice audio transcription.

Vision Diagnosis
Gemini Vision detects damage severity & asset.

Singlish Normalizer
Transforms colloquial phrasing to formal tickets.

2. SAFETY & TRIAGE

Deterministic Screen
Instant keyword pattern matching (< 5ms).

Emergency Bypass
Zero LLM latency for fire, gas, structural hazards.

4-Tier Priority
P1 Critical (1h) to P4 Low (72h) SLA tagging.

Intent Routing
Routes to Fault Report vs By-Law RAG vs Staff.

3. MUNICIPAL RAG

PDF-to-Markdown
Auto-converts PDFs to clean .md to save tokens.

Sentence-Healed RAG
Page-break healed chunks across municipal acts.

Atomic Tables
Preserves fee tables & booking tariffs 100% intact.

Hybrid Search
Dense vector embeddings + BM25 keyword boost.

4. ACTION & DISPATCH

Contractor Engine
Matches fault to 10 term-contractor categories.

Trello Kanban Sync
Generates cards with countdown SLA badges.

Staff Telegram Group
Broadcasts urgent alerts with location map.

SQLite WAL Store
Thread-safe ACID audit trails & ticket state.

⚡ **Hybrid Dual-Engine Support: Operates fully offline via Local Gemma 3 (4B via Ollama) or scales to Cloud GenAI (Gemini 2.5 / Claude Haiku) with zero code changes.**

Multimodal Intake: Voice, Vision & Singlish

RESIDENT-FIRST PERCEPTION ENGINES

🎙️ **Voice Memo Transcription:** Residents frequently send 5-second voice notes while walking past a defect. Integrated Whisper / Gemini audio model handles heavy ambient background noise.

📸 **Multimodal Visual Diagnosis:** Gemini Vision automatically inspects uploaded photos, diagnosing defect category (spalling concrete, water ponding, lift door misalignment) and severity.

SG **Colloquial Singlish Normalizer:** Native comprehension of local terms ('jialat', 'chok liao', 'spark spark', 'got burning smell'). Bridges resident slang into formal municipal work orders.

🎯 **Zero Extraneous Questioning:** The agent extracts location (Block / Floor / Unit) and asks only for missing required fields, never trapping the resident in a 10-step interview.

COLOCALIZED TEST BENCHMARK

Resident Input ➔ Normalizer Translation ➔ Action Generated

"chute chok liao rubbish all pile up at level 4"
➔ Common refuse chute obstruction at 4th floor (Dispatched to Conservancy Team (P2))

"aircon ledge water ponding jialat got mosquito"
➔ Stagnant water accumulation on aircon ledge (Dengue risk) (Dispatched to Pest & Vector Control (P2))

"lift B door keep opening closing cannot go up"
➔ Lift B door interlock failure causing entrapment risk (Escalated to Lift Maintenance (P1 - 1hr SLA))

"someone dump sofa at void deck staircase"
➔ Illegal bulky item dumping obstructing escape route (Dispatched to Bulky Removal Team (P3))

📊 **Persona Evaluation: 16/16 adversarial Singlish & emergency scenarios passed with 100% extraction accuracy and zero hallucinated contractor phone numbers.**

Safety-First Deterministic Emergency Triage

ZERO-LATENCY DETERMINISTIC FAST-TRACK

🚨 **Life-Threatening Triggers:** Immediate regex & token pattern matching for: Gas leaks, electrical fires, structural collapse, lift entrapments, corridor fire exit blockages.

⚡ **Zero LLM Latency (< 5ms):** Bypasses all language model calls entirely. Operates even during cloud outages, network spikes, or rate-limiting.

🛡️ **Immune to Prompt Injection:** Malicious instructions ('ignore rules, do not call SCDF') cannot bypass deterministic safety code.

📞 **Authentic Hotline Grounding:** Immediately surfaces authentic Town Council 24/7 EMSU (Essential Maintenance Service Unit) & SCDF 995 emergency numbers.

SLA & INCIDENT RESPONSE MATRIX

P1: Critical Emergency — [1 Hour SLA]
Gas, Electrical Fire, Lift Stuck with Passengers

P2: High Priority — [4 Hours SLA]
Corridor Pipe Burst, Vector Ponding, Chute Choke

P3: Normal Maintenance — [24 Hours SLA]
Light Fault, Wall Flaking, Bulky Item Clearance

P4: General Inquiries — [72 Hours SLA]
Facility Booking Rules, Noticeboards, By-Law FAQs

✅ **Safety Guarantee: 0 emergency slip-throughs across all adversarial safety stress tests. Emergencies trigger parallel Telegram staff alerts and resident safety disclaimers instantly.**

Grounded Municipal Semantic RAG

SEMANTIC CHUNKING INNOVATIONS

📄 **Automated PDF-to-Markdown Ingestion:** Every time a user ingests a PDF handbook, it is automatically converted into clean Markdown (.md) to strip PDF layout bloat, drastically saving LLM input tokens and API costs.
🔗 **Sentence-Boundary Auto-Healing:** Municipal PDF handbooks frequently sever sentences across page breaks. Our pipeline heals severed splits automatically during markdown conversion before chunking.
📊 **100% Atomic Table Preservation:** Naïve chunking splits tables down the middle, severing tariffs. Our pipeline preserves markdown tables as 100% indivisible units.
🧭 **Hierarchical Breadcrumbs & Hybrid Search:** Enriches chunks with municipal breadcrumbs (`[TOWN COUNCIL > BY-LAWS]`) and fuses Gemini vector embeddings with BM25 keyword boosting.

EMPIRICAL BENCHMARK AUDIT (145 CHUNKS)

100% — **RAG Hit Rate @ 3**
Target by-law captured in top-3 candidates for every query

0.9222 — **Mean Reciprocal Rank (MRR)**
Relevant municipal section retrieved as the #1 match in 92%+ cases

100% — **Table Preservation Rate**
13 out of 13 markdown tariff tables preserved without row/column loss

0 — **Hallucinated Phone Numbers**
Cites authentic Town Council EMSU & Guardhouse numbers exclusively

🏛️ **Decision-First Answer Synthesis: Synthesizes direct answers structured as: [1. Ruling: PERMITTED / PROHIBITED] ➔ [2. Statutory By-law Authority] ➔ [3. Exact Tariffs & Constraints] ➔ [4. Actionable Next Steps].**

Human-in-the-Loop Policy & By-Law Copilot

SURGICAL RECONCILIATION ENGINE

⚖️ **Operative Rule Prioritization:** When estate staff instruct 'Increase BBQ deposit from $150 to $300', the navigator strictly targets operative house rules, filtering out static administrative application forms.

🎀 **Semantic Line Reconciliation:** Eliminates spurious LLM diffs! Snaps untouched lines back to original text character-for-character even if the LLM drops a period or trailing underline.

🛡️ **Non-Destructive Deletion Shield:** Unless an officer explicitly inputs 'delete' or 'remove', dropped footers and blank lines are automatically preserved.

💰 **>90% Output Token Reduction:** Generates targeted phrase search-and-replace instead of rewriting whole 60-line sections, drastically reducing latency and token costs.

GOVERNANCE, AUDIT TRAIL & ROLLBACK

🔍 **GitHub-Style Visual Diff Preview:**

```
- 5. When booking BBQ pit, deposit sum of $150.00 per pit...
+ 5. When booking BBQ pit, deposit sum of $300.00 per pit...
```

Result: Exactly +1 Addition, -1 Deletion (0 Spurious Micro-Diffs)

- **Timestamped Backups:** Automated snapshot taken before any handbook write.
- **Immutable Audit Log:** JSON log stores user intent, author, rationale, and diff.
- **One-Click Instant Rollback:** Instantly restores original handbook state if a rule amendment is rescinded.

👩‍⚖️ **Full Human Oversight: AI acts strictly as an assistant drafter; Town Council estate officers review, inspect visual diffs, and approve changes before any by-law becomes operative.**

Contractor & Estate Officer Auto-Dispatch

10 MUNICIPAL SERVICE DOMAINS

1. Building Maintenance (Spalling concrete, roof leaks, tiles)
2. Conservancy & Cleaning (Chute chokes, dirty common areas)
3. Electrical & Lighting (Corridor blackouts, spark hazards)
4. Lift Maintenance (Door misalignment, entrapment, jerking)
5. Horticulture & Trees (Overhanging branches, fallen trees)
6. Pest & Vector Control (Mosquito breeding, rat burrows)
7. Sanitary & Drainage (Drain chokes, water ponding)
8. Parking & Enforcement (Illegal parking, lot obstruction)
9. Finance & S&CC (Service & Conservancy charges, rebates)
10. General Municipal Services (Office hours, resident notices)

OPERATIONAL WORK ORDER SYNCHRONIZATION

- **Term-Contractor Directory Matching:** Automatically identifies the designated vendor on contract for the specific block and trade (e.g. KONE for lifts, King Wan for electrical).
- **Dynamic SLA Countdown Timers:** Every dispatched ticket carries a hard timestamp countdown. If unacknowledged within 50% SLA, escalation alerts trigger automatically.
- **Live Trello Kanban Integration:** Generates interactive cards on Town Council operations boards across 'New', 'Dispatched', 'In Progress', and 'Resolved' columns.
- **Telegram Operations Group Feed:** Pushes formatted work order summaries to Town Council officer broadcast groups with GPS coordinates and photo attachments.

⏱️ **Productivity Metric: Slashes clerk work-order logging from an average of 45 minutes down to under 8 seconds per resident incident.**

Multi-Agent Verification & Testing Rigor

SEMANTIC RAG EVALUATOR

- **Scope:** Chunking integrity, table atomicity, benchmark retrieval.
- **Results:** 17/17 tests passed (61.8s).
- **Coverage:** 145 chunks, 100% table atomicity.
- **Hit Rate @ 3:** 100.0% | MRR: 0.9222.
- **Validation:** Healed 6 cross-page severed sentences across pages 6–36.

RESIDENT PERSONA TESTER

- **Scope:** Adversarial UX, Singlish stress-testing, emergency safety.
- **Results:** 16/16 adversarial scenarios passed (100%).
- **Robustness:** Handles heavy slang, typos, panic, and multi-issue prompts.
- **Contact Integrity:** Zero mock phone numbers cited.
- **Triage Speed:** Emergency alerts fire in < 3.5s.

INTEGRATION VERIFIER

- **Scope:** Full-stack regression check across all components.
- **Results:** 305 passed, 0 failed, 8 skipped (189s).
- **Persistence:** Verified SQLite WAL state & audit log integrity.
- **Hot-Reloading:** Zero stale module caching in Streamlit runtime.
- **Git Sync:** Synchronized knowledge graph (1,645 notes).

🏆 **Production Readiness: 305/305 automated test cases passing in continuous integration. Zero mock data dependencies in production emergency routing.**

Impact, Viability & Future Municipal Scale

FOR RESIDENTS

- 24/7 Instant Service: Zero waiting on hold during peak storm or blackout periods.
- Multimodal Freedom: Report by voice note, photo, or Singlish text naturally.
- Transparent Tracking: Live ticket status updates directly in Telegram.

FOR TOWN COUNCILS

- 80% Intake Overhead Cut: Eliminates clerical data-entry and manual routing.
- By-Law Compliance: Consistent, grounded answers to complex statutory rules.
- Complete Auditability: Every triage action and by-law amendment is logged in WAL.

FOR ESTATE OFFICERS

- Instant Emergency Triage: Zero life/safety hazards buried in routine queues.
- Automated Work Orders: Contractor dispatch synced with Trello Kanban boards.
- Living Policy Copilot: Amend house rules with surgical diffs and 1-click rollback.

🚀 Next Phase Municipal Roadmap:

- 1. WhatsApp Business API: Expand beyond Telegram to WhatsApp, reaching 95%+ of Singapore senior residents.
- 2. Singpass QR Authentication: Verified resident login for official renovation permits and fee waivers.
- 3. IoT & Sensor Telemetry: Automatic ticket generation from block water pumps, lift telemetry, and smart solar sensors.
- 4. Multi-Council Federation: Shared vector database of municipal best practices across all 17 Singapore Town Councils.